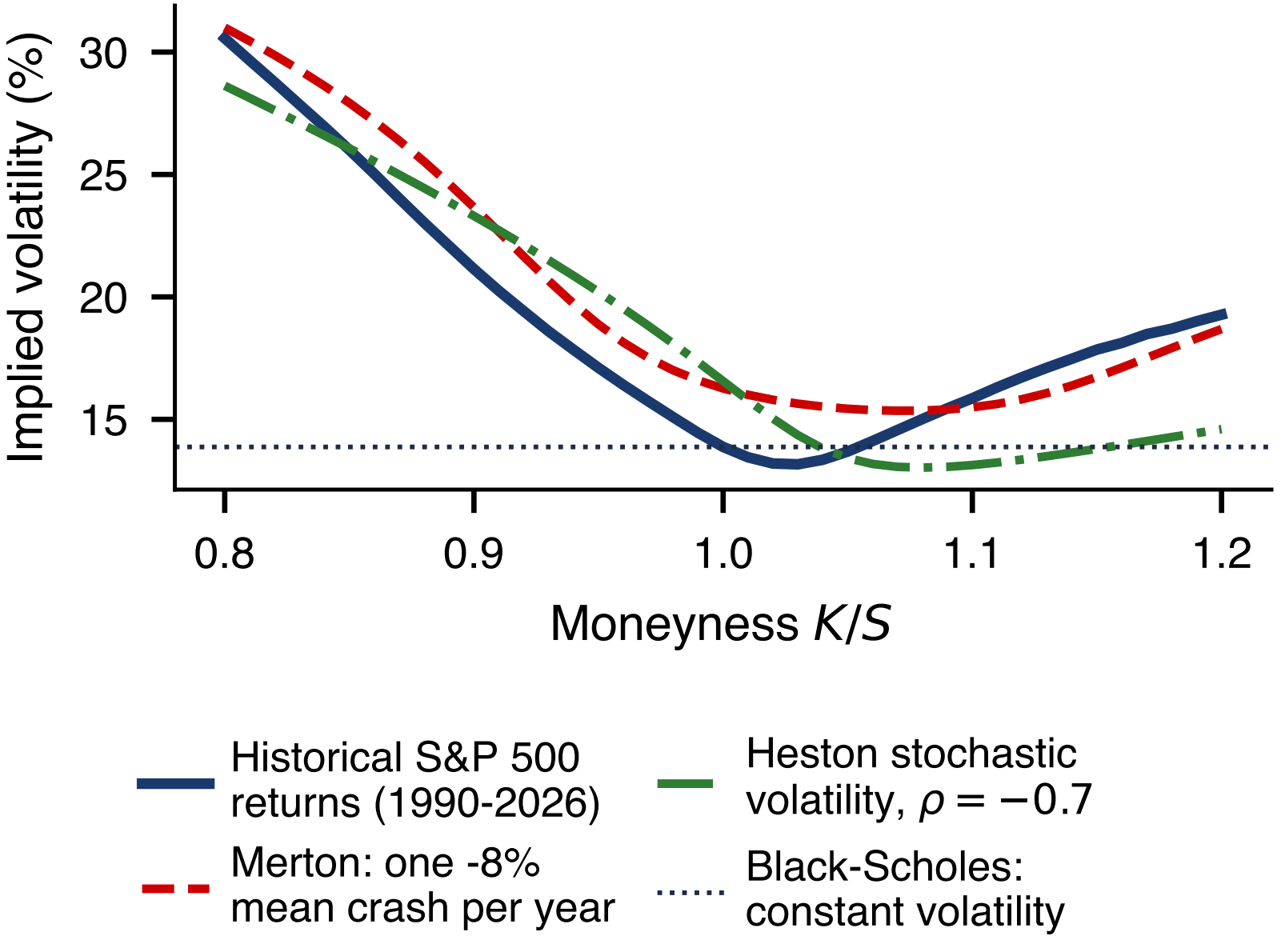
Implied volatility (%)
30
25
20
15
0.8
0.9
1.0
1.1
1.2
Moneyness K/S
Historical S&P 500 returns (1990-2026)
Merton: one -8% mean crash per year
Heston stochastic volatility, ρ = −0.7
Black-Scholes: constant volatility